

Introduction to Sorting Algorithms

Sorting rearranges the elements of a list into a defined order.

Comparison-based sorts are bounded below by $O(n \log n)$ worst case.

Stability means equal elements keep their original relative order.

Merge Sort

Merge sort splits the input in half recursively.

Each half is sorted independently, then merged in linear time.

Merge sort is stable and runs in $O(n \log n)$ time in all cases.

Quick Sort

Quick sort partitions the array around a chosen pivot element.

Average case running time is $O(n \log n)$; worst case is $O(n^2)$.

In place partitioning avoids the extra memory merge sort needs.

Summary

Choose merge sort when stability or worst-case guarantees matter.

Choose quick sort when average-case speed and low memory matter.